

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Supplementary Winter Examination – 2023**

Branch: Electronics and Communication Engineering / Electronics and Telecommunication Engineering

Course: B. Tech

Sem.:- VII

Subject with Subject Code: - Digital communication BTETC701_Y22

Marks: 60

Date:-02-01-2024

Time:3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Explain the basic digital communication system with neat block diagram.	L1	6
B) What are the various types of errors introduced during transmission? Explain.	L2	6
C) Explain the operation of PCM generator with neat block diagram. Also discuss uniform quantization, quantization error and transmission bandwidth of PCM.	L2	6
Q.2 Solve Any Two of the following.		12
A) Explain the working of DEPSK system with the help of transmitter receiver block diagrams. Also state the advantages and disadvantages of the same.	L2	6
B) Write a note on Ergodic processes.	L1	6
C) Write a note on M-ary PSK.	L3	6
Q.3 Solve Any Two of the following.		12
A) Explain the working of QPSK receiver with neat block diagram. Also explain the carrier synchronization in QPSK.	L3	6
B) Explain the working of optimum receiver system with neat block diagram. Also obtain equation for probability of error for the same.	L2	6
C) What are the cause and effects of ISI? Explain Eye diagram.	L2	6

Q.4 Solve Any Two of the following. 12

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| A) Write a note on narrowband noise. Give representation of narrowband noise in terms of in phase & quadrature components. | L2 | 6 |
| B) Explain T-Carrier system. | L1 | 6 |
| C) What is Matched Filter? Explain its working with the help of neat block diagram. | L3 | 6 |

Q. 5 Solve Any Two of the following. 12

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| A) Explain the concept of Jammers. What are its applications? | L1 | 6 |
| B) Explain the working of Pseudo noise sequences generator. | L2 | 6 |
| C) Explain Frequency-Hopping Spread Spectrum technique. What are its advantages and applications? | L3 | 6 |

*** End ***